THE CHINESE UNIVERSITY OF HONG KONG
2026-27 Term 1 AISC5020 Mathematics & Statistics for AI
Homework Assignment 1
Due Date: 11 October 2026 (Sunday), 11:59PM

Full marks: 60

Instructions: Answer all questions and submit your solutions together with the signed declaration via Gradescope.

I declare that: (i) the assignment here submitted is original except for source material explicitly acknowledged; and (ii) the piece of work, or a part of the piece of work has not been submitted for more than one purpose (e.g. to satisfy the requirements in two different courses) without declaration. I also acknowledge that I am aware of University policy and regulations on honesty in academic work, and of the disciplinary guidelines and procedures applicable to breaches of such policy and regulations, as contained on the University website https://www.cuhk.edu.hk/policy/academichonesty/

I declare that I have not distributed/shared/copied any teaching materials without the consent of the course teacher(s) to gain unfair academic advantage in the assignment.

I declare that I have read and understood the University's policy on the use of AI for academic work. I confirm that I have complied with the instructions given by my course teacher(s) regarding the use of AI tools for this assignment and consent to the use of AI content detection software to review my submission.

I also understand that assignments without a properly signed declaration by the student concerned will not be graded by the teacher(s).

Signature	Date
Name	Student ID

1. **[10 marks] Vectors, dot products, norms, distances, and orthogonality**

   Let

   $$u = (3, 4), \qquad v = (4, -3), \qquad w = (1, 2).$$

   Calculate the dot products

   $$u \cdot v, \qquad u \cdot w, \qquad v \cdot w,$$

   the norms

   $$\|u\|_2, \qquad \|v\|_2, \qquad \|w\|_2,$$

   and the distances

   $$d(u, v) = \|u - v\|_2, \qquad d(u, w) = \|u - w\|_2, \qquad d(v, w) = \|v - w\|_2.$$

   Which pair(s) of vectors among u, v, w are orthogonal? Explain briefly.

2. **[10 marks] Linear independence and span**

   Let

   $$v_1 = (1, 2, 0), \qquad v_2 = (2, 4, -1), \qquad v_3 = (0, 0, 1).$$

   (a) Determine whether the set $\{v_1, v_2, v_3\}$ is linearly independent. Justify your answer.

   (b) Describe the span $\text{span}\{v_1, v_2, v_3\}$ as simply as possible.

3. **[10 marks] Eigen-decomposition and singular value decomposition**

   (a) Let

   $$A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 2 \end{pmatrix}.$$

   Find the eigenvalues of A. Then, find a unit eigenvector for each eigenvalue, and write an orthogonal eigen-decomposition

   $$A = QDQ^T,$$

   where the columns of Q are orthonormal eigenvectors and D is a diagonal matrix of eigenvalues.

   (b) Let

   $$B = \begin{pmatrix} 3 & 0 \\ 0 & 2 \\ 0 & 0 \end{pmatrix}.$$

   Find the singular values of B and write a full singular value decomposition

   $$B = USV^T$$

   with $U \in \mathbb{R}^{3\times3}$, $S \in \mathbb{R}^{3\times2}$, and $V \in \mathbb{R}^{2\times2}$.

   Hint: The two columns of B are already orthogonal.

4. **[10 marks] Graph Laplacian and spectral quantities**

Consider the undirected, unweighted graph with nodes $\{v_1, v_2, v_3, v_4\}$ and edges

$$\{v_1, v_2\}, \quad \{v_1, v_3\}, \quad \{v_1, v_4\}.$$

(a) Write down the adjacency matrix A and the degree matrix D. Then compute the Laplacian matrix

$$L = D - A.$$

(b) The eigenvalues of L are

$$0, \quad 1, \quad 1, \quad 4.$$

State the algebraic connectivity of the graph, the largest Laplacian eigenvalue, and determine whether the graph is connected. Briefly explain how the number of zero eigenvalues is related to connected components.

5. **[10 marks] Multivariable calculus**

(a) Let

$$f(x, y, z) = x^2 + \sin(xy) + 2x + \cos(z) + 3.$$

Calculate the gradient ∇f and the Hessian matrix H_f at $(0, 0, 0)$. Then write the second-order Taylor approximation of f at $(0, 0, 0)$.

(b) Let

$$F(x, y, z) = \begin{pmatrix} x + y + z \\ xyz \end{pmatrix}.$$

Calculate the Jacobian matrix $J_F(x, y, z)$, and evaluate it at $(x, y, z) = (1, 2, 3)$.

6. **[10 marks] Automatic differentiation**

Consider the simple computation graph

$$z = wx + b, \qquad L = (z - t)^2,$$

where x and t are fixed inputs. Suppose

$$w = 2, \qquad x = 3, \qquad b = 1, \qquad t = 5.$$

(a) Perform the forward pass: compute the values of z and L.

(b) Using reverse-mode automatic differentiation, compute

$$\frac{\partial L}{\partial z}, \qquad \frac{\partial L}{\partial w}, \qquad \frac{\partial L}{\partial b}.$$